



Magnetic Properties 2

Presented by Dr Adijat 'Wumi Inyang

Magnetic behaviour

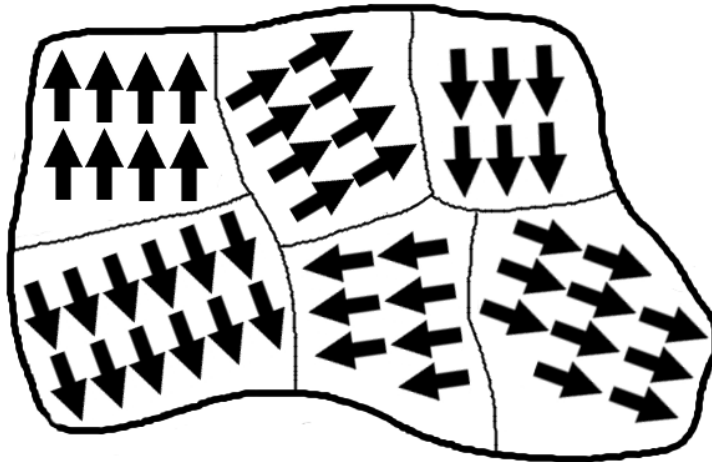
Different materials require different levels of magnetism to perform their functions

- Magnets in generator is required to be a hard magnetization to maintain the generation of a high magnetic field
- Magnets in a computer hard disk or storage hard drive, the magnetism required is sufficient for performing their storage function but soft enough for allowing re-writing



Magnetic behaviour of iron

- Magnetic behavior of α -Fe (ferrite)



Single Crystal

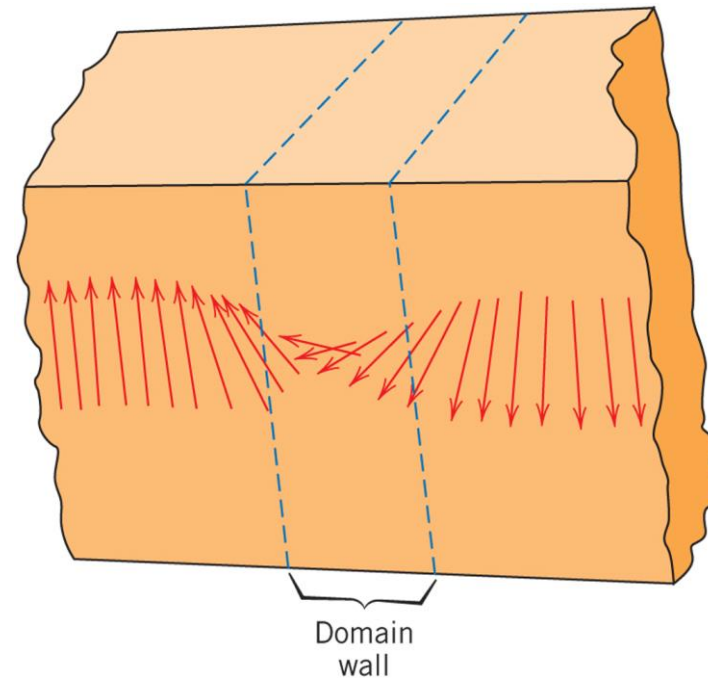
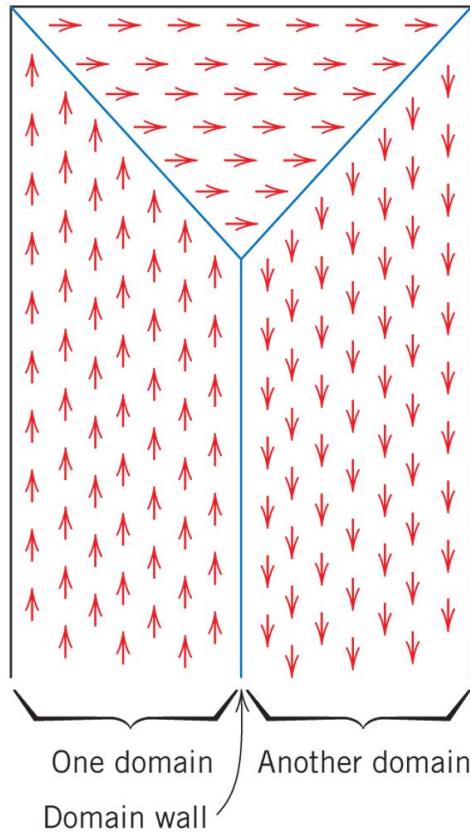
Net magnetic moment is 0

Domain: regions within a crystal that have magnetic moments aligned in the same direction



Magnetic behaviour

- **Domains** in ferromagnetic or ferrimagnetic materials



In polycrystalline materials each grain may consist of more than a single domain

Magnetic behaviour

- **Domains** in ferromagnetic or ferrimagnetic materials

As the applied field (H) increases the magnetic domains change shape and size by movement of domain boundaries.

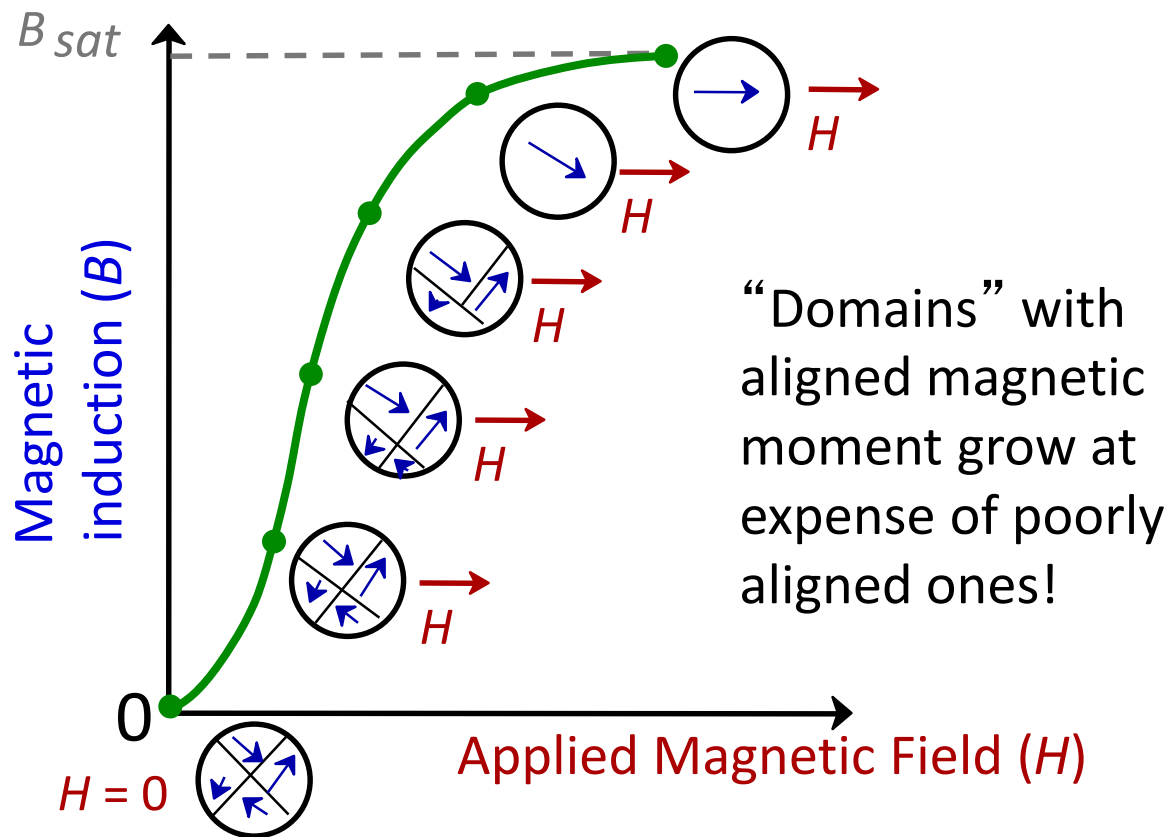
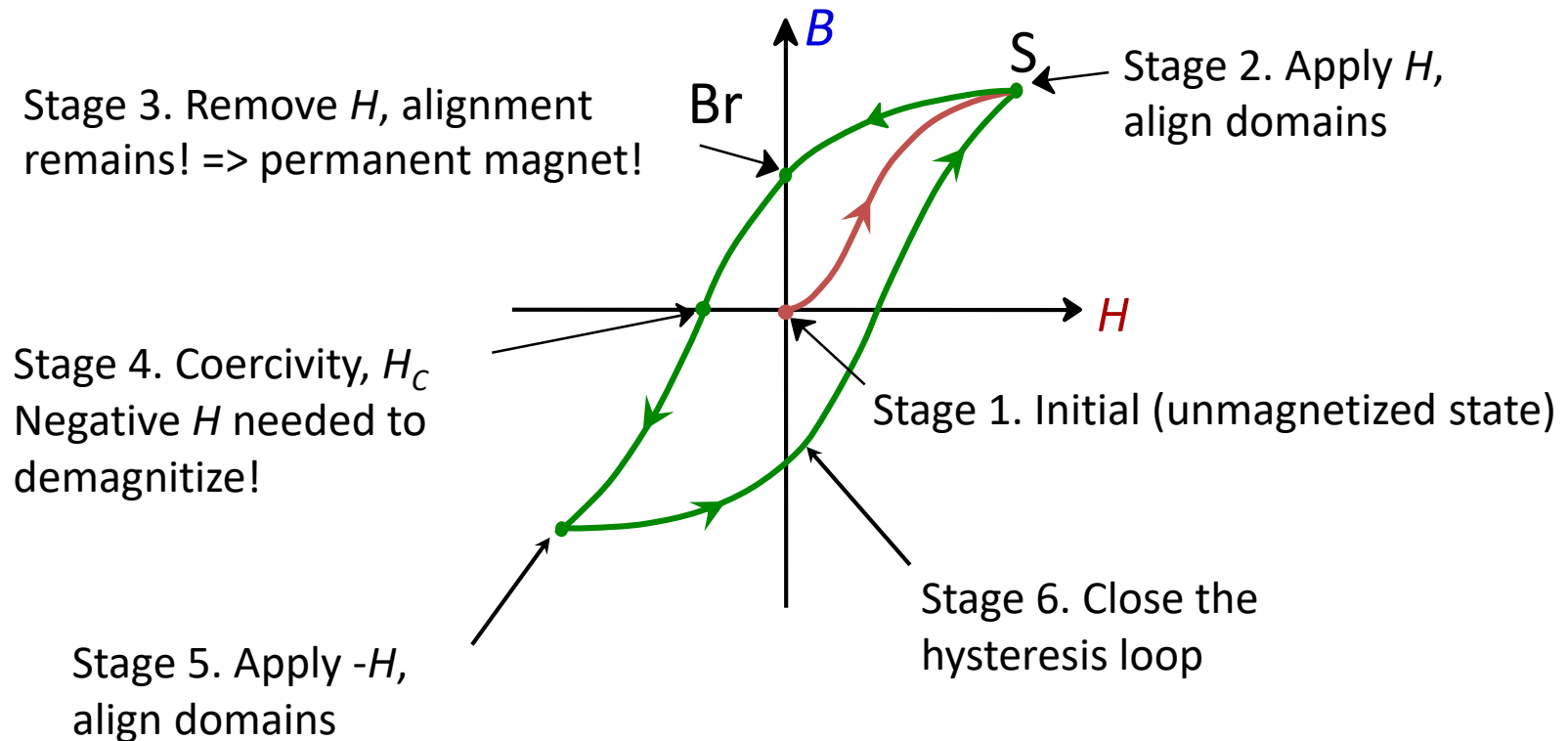


Fig. 20.13, Callister & Rethwisch 9e.

Magnetic behaviour

- **Hysteresis** and permanent magnetization

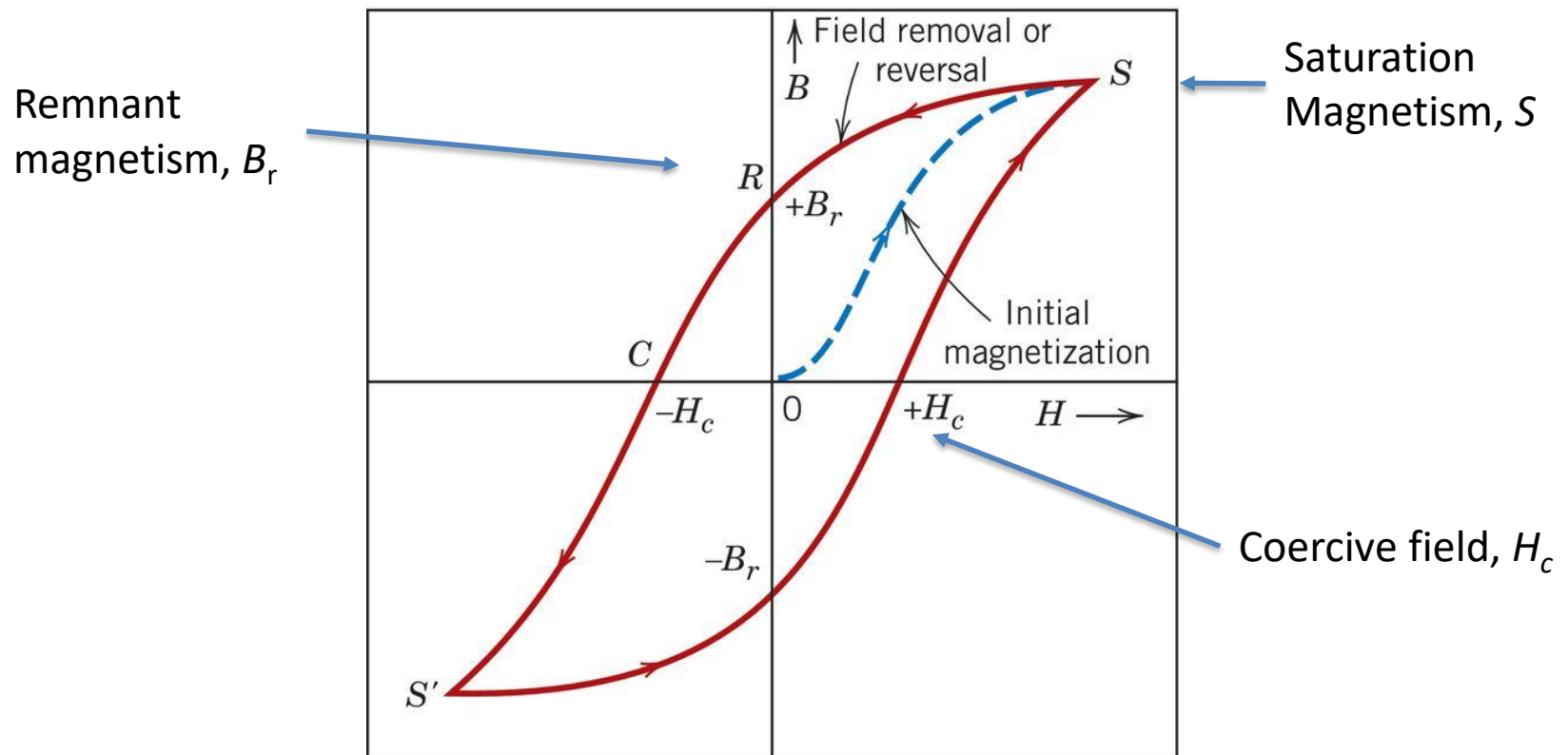
The magnetic hysteresis phenomenon



Magnetic behaviour

- **Hysteresis** and permanent magnetization

The magnetic hysteresis phenomenon



Magnetic behaviour

- Hysteresis in **Hard and Soft Magnetic Materials**

Hard magnetic materials:

- large coercivities
- used for permanent magnets
- add particles/voids to inhibit domain wall motion
- example: tungsten steel --
 $H_c = 5900$ amp-turn/m)

Soft magnetic materials:

- small coercivities
- used for electric motors
- example: commercial iron 99.95 Fe

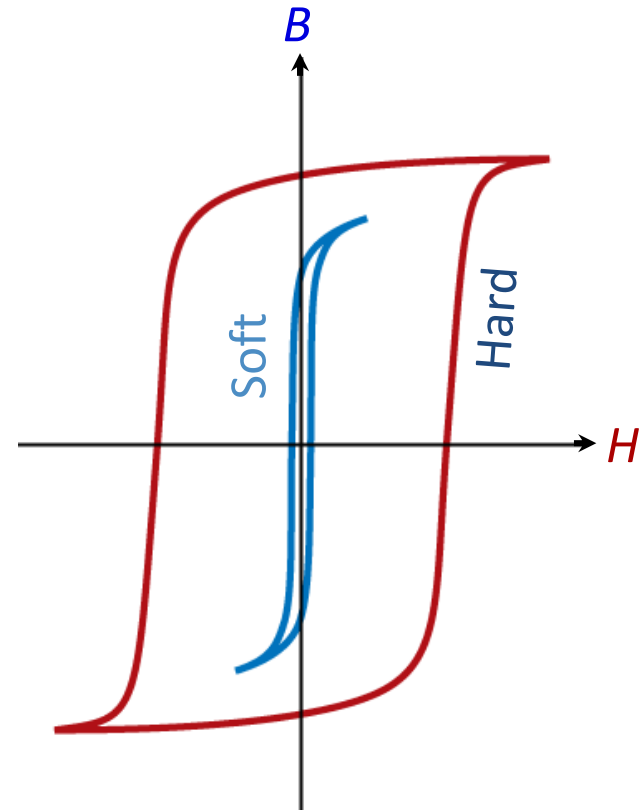
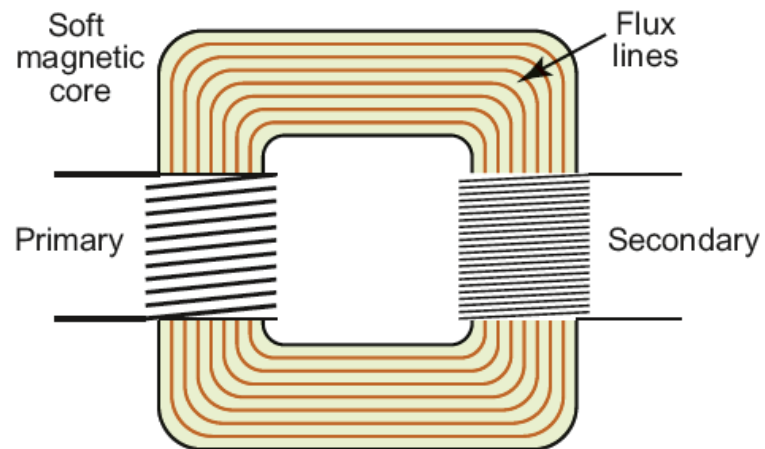


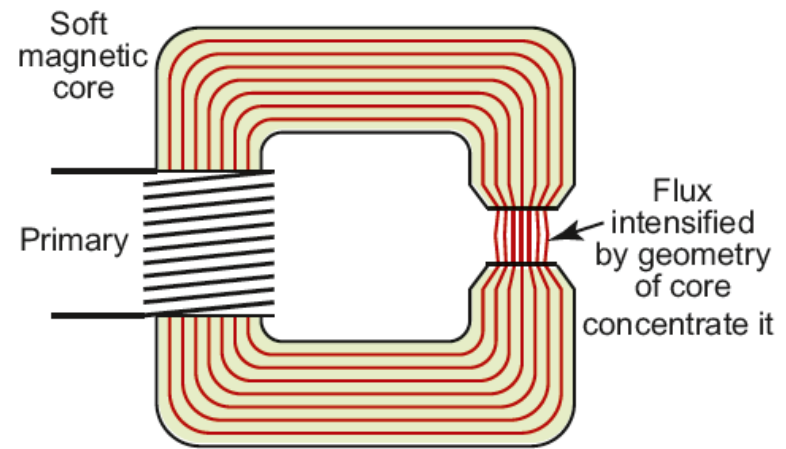
Fig. 20.19, Callister & Rethwisch 9e.

Soft magnetic devices

- Devices such as **transformers** and **electromagnets** must magnetize easily when subjected to an applied field, yet lose their magnetism when the field is switched off
- Soft magnetic materials are used to conduct and focus the magnetic flux



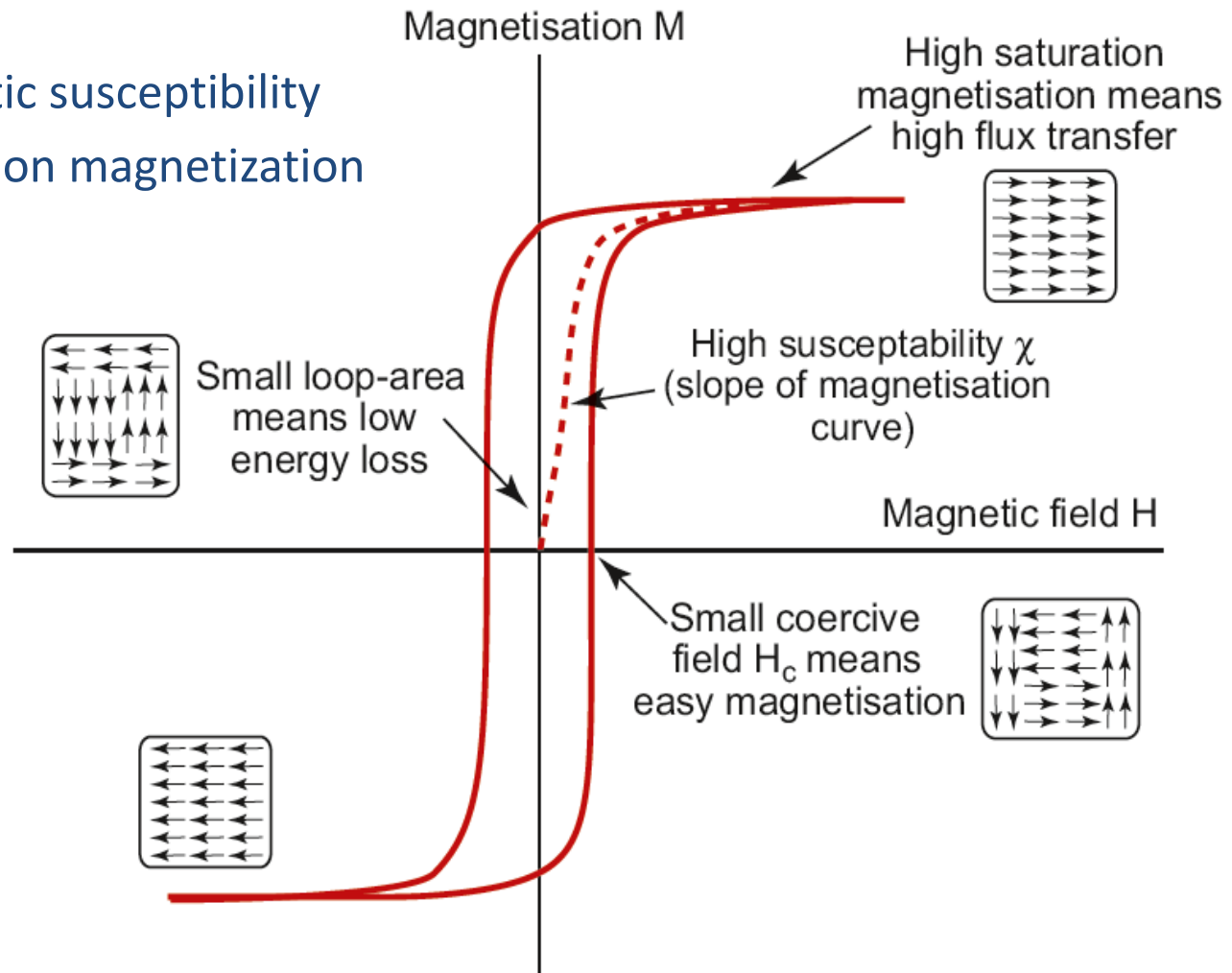
Transformer



Electromagnet

Soft magnetic devices

- High magnetic susceptibility
- High saturation magnetization



Magnetic material properties

Soft magnetic materials

<i>Material</i>	<i>Composition (wt%)</i>	<i>Initial Relative Permeability μ_i</i>	<i>Saturation Flux Density B_s [tesla (gauss)]</i>	<i>Hysteresis Loss/Cycle [J/m³ (erg/cm³)]</i>	<i>Resistivity ρ ($\Omega \cdot m$)</i>
Commercial iron ingot	99.95 Fe	150	2.14 (21,400)	270 (2,700)	1.0×10^{-7}
Silicon-iron (oriented)	97 Fe, 3 Si	1,400	2.01 (20,100)	40 (400)	4.7×10^{-7}
45 Permalloy	55 Fe, 45 Ni	2,500	1.60 (16,000)	120 (1,200)	4.5×10^{-7}
Supermalloy	79 Ni, 15 Fe, 5 Mo, 0.5 Mn	75,000	0.80 (8,000)	—	6.0×10^{-7}
Ferroxcube A	48 MnFe ₂ O ₄ , 52 ZnFe ₂ O ₄	1,400	0.33 (3,300)	~40 (~400)	2,000
Ferroxcube B	36 NiFe ₂ O ₄ , 64 ZnFe ₂ O ₄	650	0.36 (3,600)	~35 (~350)	10^7

Table 21.5, Callister & Rethwisch 9e..

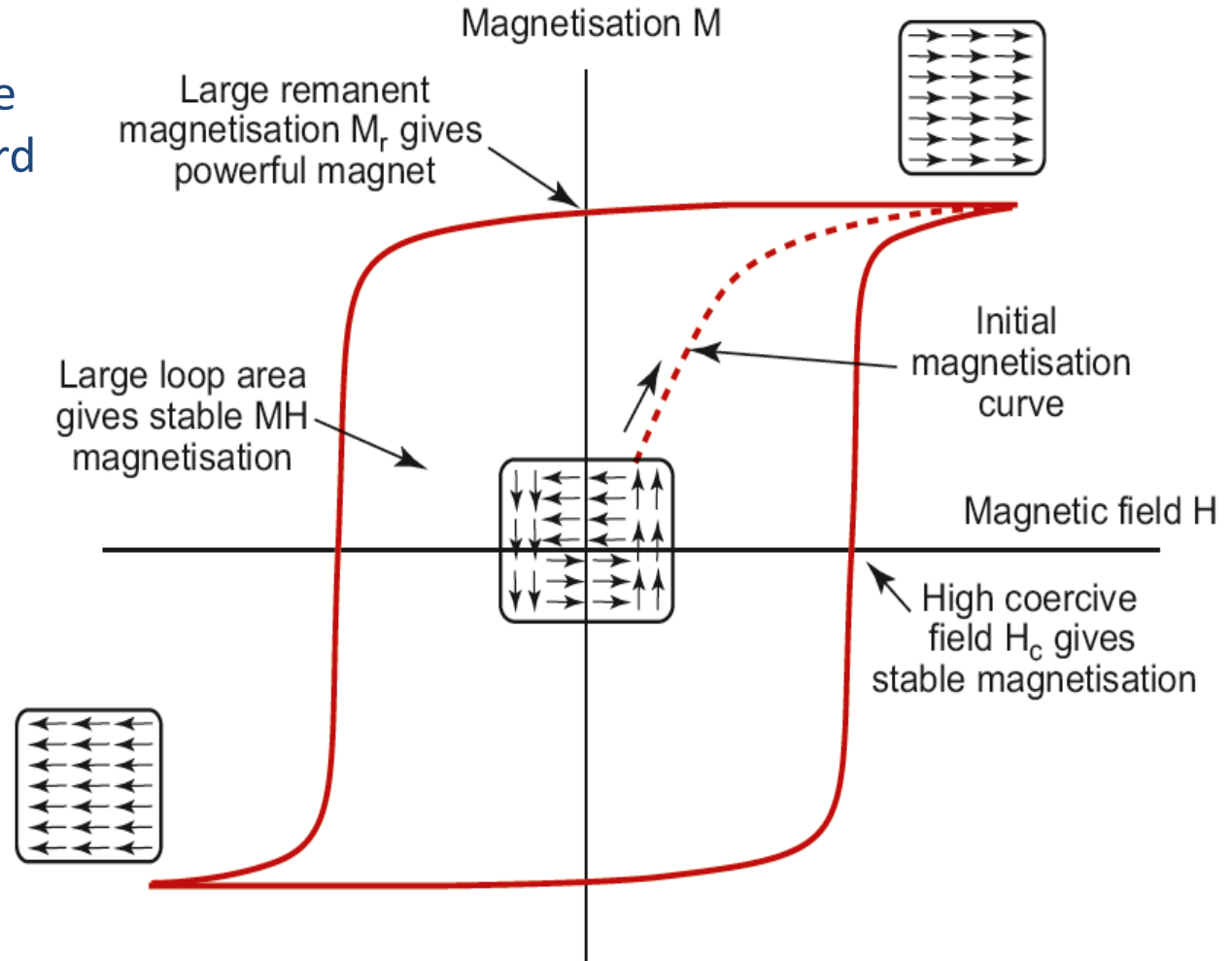
Soft magnetic devices

- Materials for soft magnetic applications

Application	Frequency f	Material requirements	Material choice
Electro-magnets	< 1 Hz	High M_s , high χ	Silicon iron Fe–Co alloys (Permendur)
Motors, low-frequency transformers	< 100 Hz	High M_s , high χ	Silicon-iron AMA (amorphous Ni–Fe–P alloys)
Audio amplifiers, loudspeakers, microphones	< 100 kHz	High M_s , very low H_c , and high χ	Ni–Fe (Permalloy, Mumetal)
Microwave and UHF applications	$< \text{MHz}$	High M_s , very low H_c , electrical insulator	Cubic ferrites: MFe_2O_4 with $\text{M} = \text{Cu} / \text{Mn} / \text{Ni}$
Giga-hertz devices	> 100 MHz	Ultra-low hysteresis, excellent insulator	Garnets

Hard magnetic devices

- High remanence magnetization is the key property for hard magnets
- Examples include magnetic clutches and earphones



Magnetic material properties

Hard magnetic materials

<i>Material</i>	<i>Composition (wt%)</i>	<i>Remanence B_r [tesla (gauss)]</i>	<i>Coercivity H_c [amp- turn/m (Oe)]</i>	<i>$(BH)_{max}$ [kJ/m³ (MGOe)]</i>	<i>Curie Temperature T_c (°C)</i>	<i>Resistivity ρ ($\Omega \cdot m$)</i>
Tungsten steel	92.8 Fe, 6 W, 0.5 Cr, 0.7 C	0.95 (9,500)	5,900 (74)	2.6 (0.33)	760	3.0×10^{-7}
Cunife	20 Fe, 20 Ni, 60 Cu	0.54 (5,400)	44,000 (550)	12 (1.5)	410	1.8×10^{-7}
Sintered alnico 8	34 Fe, 7 Al, 15 Ni, 35 Co, 4 Cu, 5 Ti	0.76 (7,600)	125,000 (1,550)	36 (4.5)	860	—
Sintered ferrite 3	BaO–6Fe ₂ O ₃	0.32 (3,200)	240,000 (3,000)	20 (2.5)	450	$\sim 10^4$
Cobalt rare earth 1	SmCo ₅	0.92 (9,200)	720,000 (9,000)	170 (21)	725	5.0×10^{-7}
Sintered neodymium– iron–boron	Nd ₂ Fe ₁₄ B	1.16 (11,600)	848,000 (10,600)	255 (32)	310	1.6×10^{-6}

Table 21.6, Callister & Rethwisch 9e..

Hard magnetic devices

- Materials for hard magnetic applications

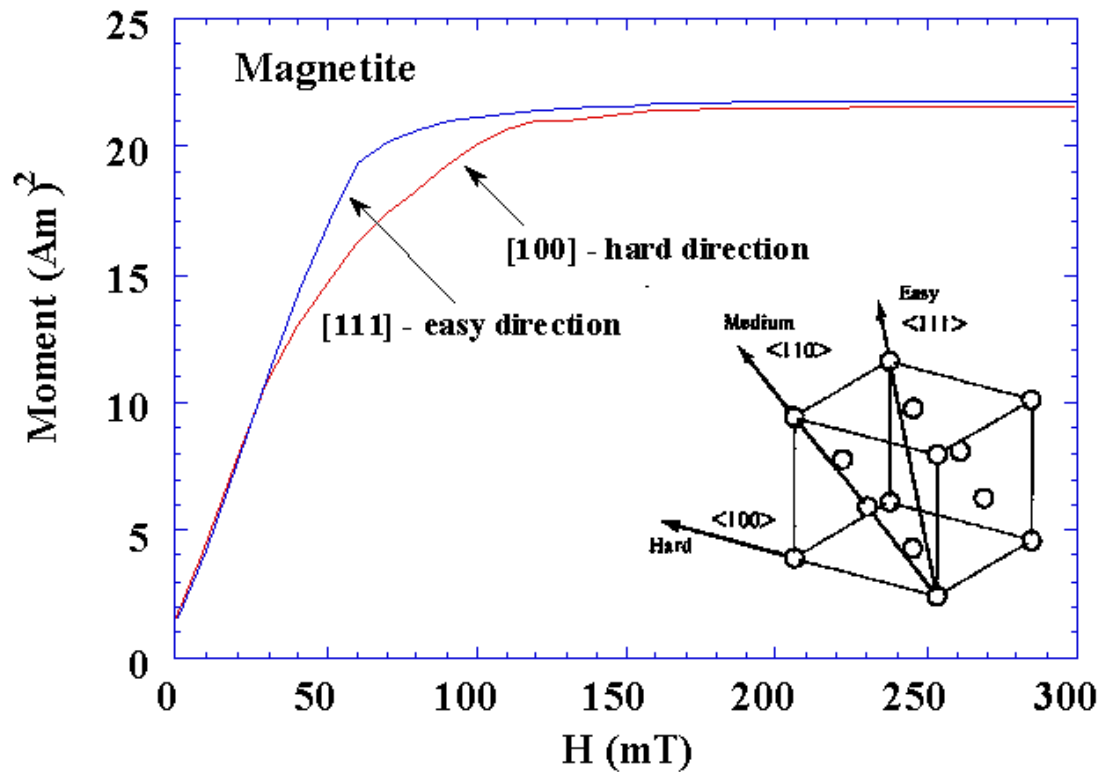
Application	Material requirements	Material choice
Large permanent magnets	High M_r , high H_c , and high-energy product $(MH)_{\max}$; low cost	Alnicos Hexagonal ferrites
Small high-performance magnets	Acoustic, electronic and miniature mechanical applications, electric propulsion, wind turbine generators	Cobalt-samarium Neodymium-iron-boron (Platinum-cobalt)
Information storage	Thin, 'square' hysteresis loop	Elongated particles of Fe_2O_3 , Cr_2O_3 or hexagonal ferrite

Magnetic anisotropy

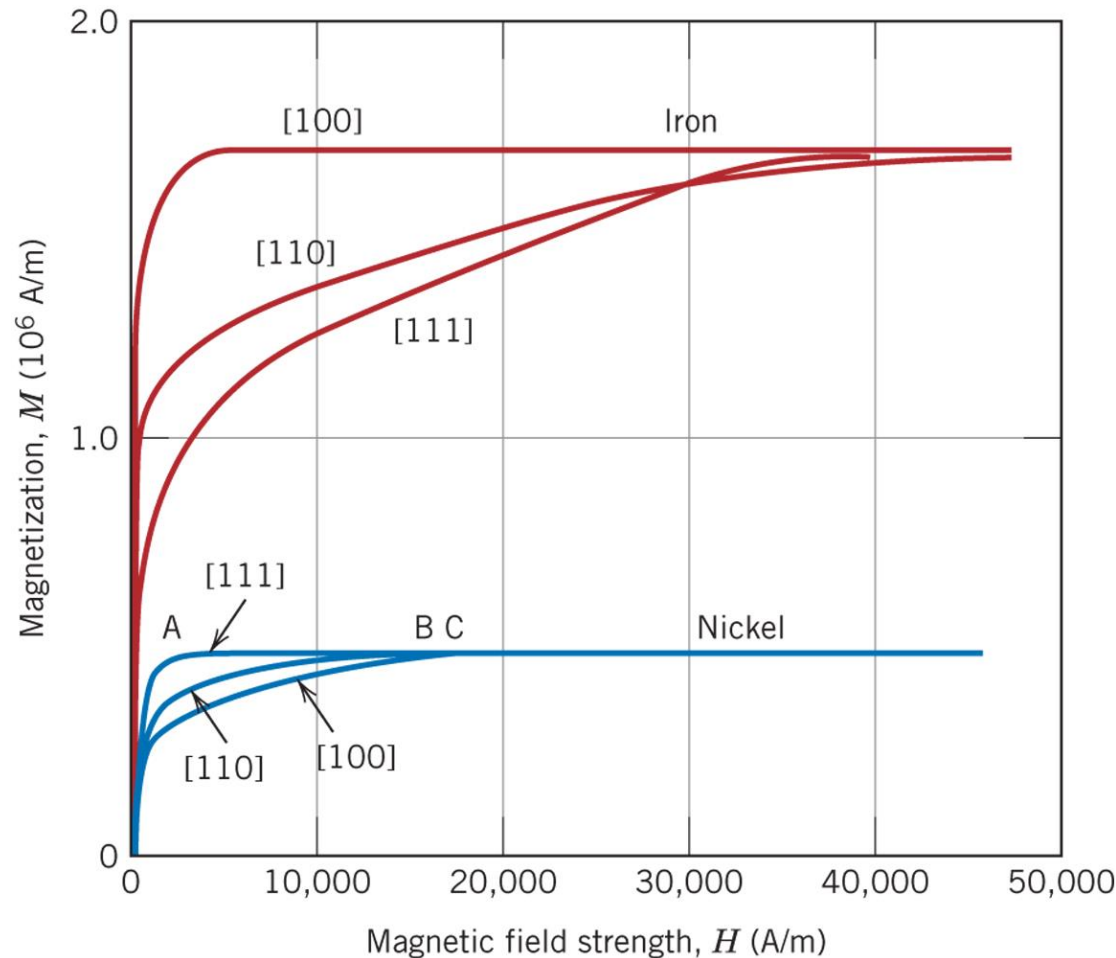
Dependence of magnetic behavior on the crystallographic orientation of a material.

It is influenced by:

- (1) whether the specimen is a single or polycrystal
- (2) if polycrystalline, any preferred orientation of the grains
- (3) The presence of second-phase particles
- (4) other factors such as temperature or mechanical stress



Magnetic anisotropy Fe and Ni



Magnetization curves for single crystals of iron and nickel. For both metals, a different curve was generated when the magnetic field was applied in each of [100], [110], and [111] crystallographic directions.

[Adapted from Callister, Chapter 20.5]

Magnetic storage media

- Hard disk drives (**granular/perpendicular media**):

- CoCr alloy grains (darker regions) separated by oxide grain boundary segregant layer (lighter regions)
- Magnetization direction of each grain is perpendicular to plane of disk

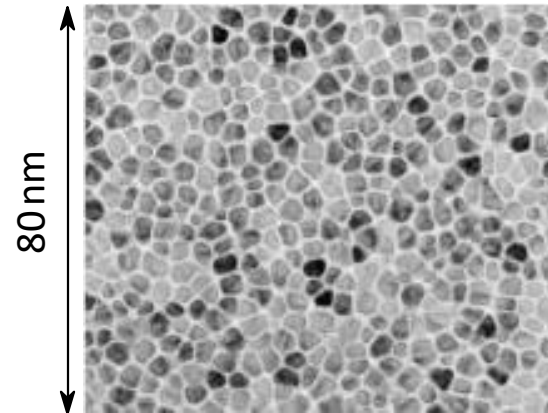
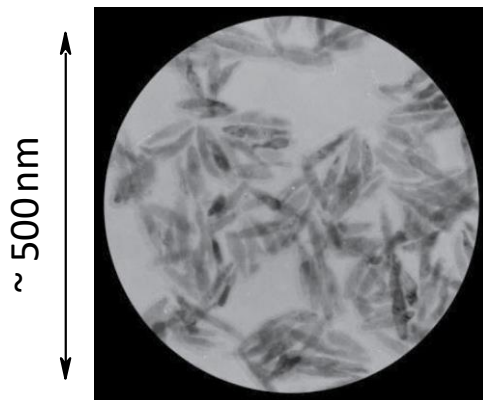
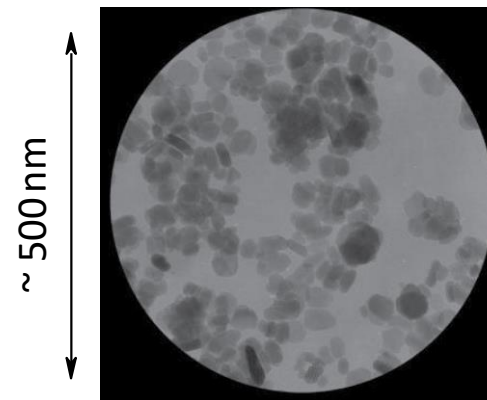


Fig. 20.24, Callister & Rethwisch 9e.

- Recording tape (**particulate media**):



- Acicular (needle-shaped) ferromagnetic metal alloy particles



- Tabular (plate-shaped) ferrimagnetic barium-ferrite particles

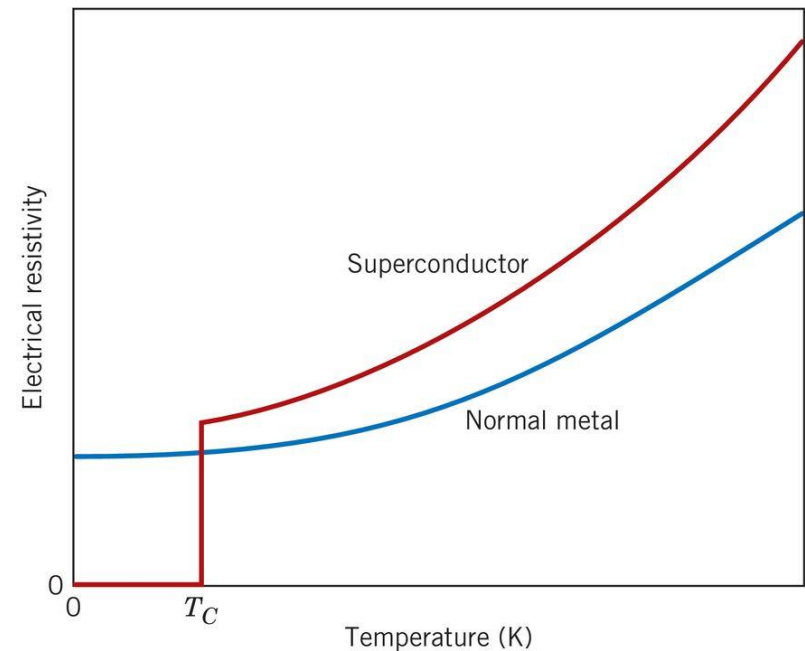
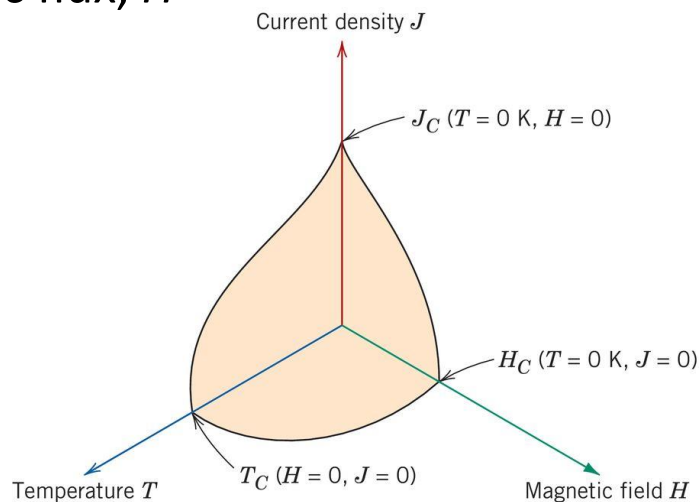
Fig. 20.25, Callister & Rethwisch 9e.

Superconductivity

Superconductivity is an electrical conduction effect.

Electrical resistivity reduces with temperature for most high purity metals
Some materials the resistivity suddenly drops to zero at a temperature called the **critical temperature**, T_C

The change to superconductivity also depends on the current density, J , and magnetic flux, H



Superconductivity



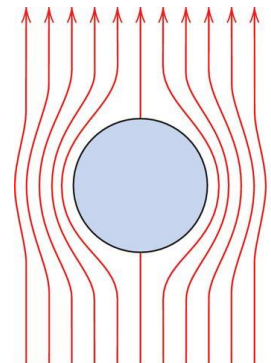
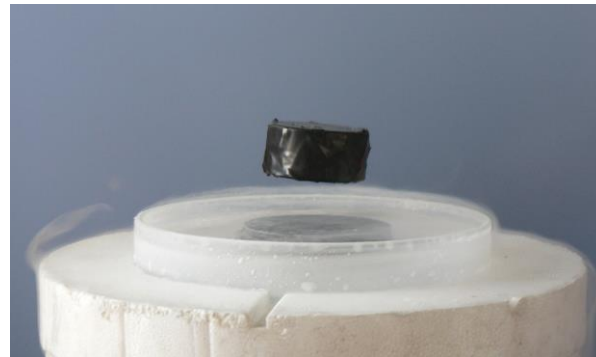
Superconductivity is an electrical conduction effect.

In the superconductive state, attractive interactions between pairs of conducting electrons make the usual interference by phonons ineffective – creating a low resistivity condition

(Note: this is not the same as the quantum conduction in a carbon nanotube)

The resulting diamagnetic response can be classified as:

Type 1: completely diamagnetic, in which the magnetic field is excluded the material (the *Meissner effect*)



Type 2: are diamagnetic at low magnetic fields but then gradually change to normal conductive behavior as the magnetic field flux increases

Superconductivity

<i>Material</i>	<i>Critical Temperature T_C (K)</i>	<i>Critical Magnetic Flux Density B_C (tesla)^a</i>
Elements^b		
Tungsten	0.02	0.0001
Titanium	0.40	0.0056
Aluminum	1.18	0.0105
Tin	3.72	0.0305
Mercury (α)	4.15	0.0411
Lead	7.19	0.0803
Compounds and Alloys^b		
Nb–Ti alloy	10.2	12
Nb–Zr alloy	10.8	11
PbMo ₆ S ₈	14.0	45
V ₃ Ga	16.5	22
Nb ₃ Sn	18.3	22
Nb ₃ Al	18.9	32
Nb ₃ Ge	23.0	30
Ceramic Compounds		
YBa ₂ Cu ₃ O ₇	92	—
Bi ₂ Sr ₂ Ca ₂ Cu ₃ O ₁₀	110	—
Tl ₂ Ba ₂ Ca ₂ Cu ₃ O ₁₀	125	—
HgBa ₂ Ca ₂ Cu ₂ O ₈	153	—

**High temperature
Superconductors
(> 77K liq N₂)**

Summary



Magnetic behaviour:

- Hysteresis loops
- Hard and soft magnetic materials
- Applications of magnetic materials
- Superconductivity